Dark Energy as an Epistemological Artifact: A Geometric and Wave–Functional Alternative on the Compact 3–Sphere (S^3) to Dynamical Dark Energy

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(Dated: 2025)

Recent DESI observations and the work of Ishiyama, Prada, and Klypin [*Phys. Rev. D* **112**, 043504 (2025)] suggest time–varying (dynamical) dark energy as an extension of ΛCDM. However, the very concept of dark energy lacks ontological and epistemological grounding. We present a mathematically complete alternative: the Universe's late acceleration emerges from the geometry and phase structure of a compact three–manifold—the 3–sphere S^3—without any repulsive "energy." The model unifies a golden–ratio self–similar pre–recombination regime with an exponential post–recombination phase, yielding the observed acceleration as a purely geometric invariant. We provide closed–form equations, continuity relations, and parameter links, address curvature constraints, and compare to DESI BAO, Planck CMB spectra, and Pantheon+ supernovae. For PRD rigor we include MCMC analyses with full chains, CAMB spectra, and joint DESI+SN+CMB fits.

I. INTRODUCTION

The ΛCDM paradigm encodes acceleration via a hypothetical component termed "dark energy." No physical ontology, interaction mechanism, or independent detection firmly establishes such an entity. We show that the same acceleration arises from the internal geometry and harmonic phase of a simply connected compact 3–manifold, S^3 with trivial fundamental group. We expand derivations, confront DESI Year 1 BAO, Planck 2018 CMB power spectra, and Pantheon+ distances, and address curvature bounds. Full MCMC chains and CAMB spectra are referenced in the Appendices.

II. WAVE ONTOLOGY ON S^3

A. Total wave function

Consider a scalar on the curved background obeying the Klein–Gordon equation

$$(\Box + m^2)\,\Psi = 0 \tag{1}$$

with spatial Laplace–Beltrami eigenvalues on S^3 of radius R given by $-n(n+2)/R^2$ with multiplicity $(n+1)^2$, $n \in \mathbb{N}_0$. We represent the state as a two–phase superposition,

$$\Psi(t) = \Psi_{\rm m}(t) + \Psi_{\rm v}(t) \equiv A(t)\, e^{i\phi(t)}, \tag{2}$$

with normalization

$$|\Psi_{\rm m}|^2 + |\Psi_{\rm v}|^2 = 1 \;\Rightarrow\; |\Psi_{\rm m}|^2 = \cos^2\Theta(t), \quad |\Psi_{\rm v}|^2 = \sin^2\Theta(t). \tag{3}$$

A microscopic route uses a Gross–Pitaevskii functional on S^3 with phase coupling; see App. A.

B. Phase as driver of expansion

Let the instantaneous phase–frequency difference define

$$\dot\phi(t) \equiv \omega_{\rm m}(t) - \omega_{\rm v}(t) \equiv H(t). \tag{4}$$

In the post–recombination regime where the void–like phase dominates and the spectrum geometrically locks, $H(t) \approx H = \text{const}$.

III. POST–RECOMBINATION EXPONENTIAL PHASE

A. Wave function and scale factor

For constant H the time mode yields

$$\psi_{\rm post}(t) = B\,\exp\!\left[-\frac{3H}{2}(t - t_{\rm rec})\right] e^{iHt}, \qquad t \geq t_{\rm rec}. \tag{5}$$

The S^3 radius (scale factor) obeys

$$R(t) = R_{\rm rec}\, e^{H(t-t_{\rm rec})}, \qquad a(t) \equiv \frac{R(t)}{R_{\rm rec}} = e^{H(t-t_{\rm rec})}. \tag{6}$$

B. Intrinsic acceleration and invariant

From (6):

$$\dot R = HR, \qquad \ddot R = H^2 R \quad \Rightarrow \quad \frac{\ddot a}{a} = H^2 > 0. \tag{7}$$

In GR variables this corresponds to an *effective* cosmological invariant

$$\Lambda_{\rm eff} = \frac{3H^2}{c^2} \tag{8}$$

emerging from topology/phase rather than a vacuum fluid.

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IV. FRACTAL PRE–RECOMBINATION PHASE

A. Golden–ratio radius law

Prior to recombination we posit self–similar growth on S^3:

$$R(t) = \ell_0\, \phi^{N(t)}, \qquad N(t) = \left\lfloor \log_\phi\!\left(\frac{ct}{\ell_0}\right) \right\rfloor. \tag{9}$$

B. Matching wave mode

A matching amplitude consistent with $V(t) = 2\pi^2 R^3(t)$ is

$$\psi_{\rm pre}(t) = A\left(\frac{ct}{\ell_0}\right)^{-3/2} e^{i\phi t/t_\ell}, \qquad t_\ell \equiv \ell_0/c, \tag{10}$$

so that $|\psi_{\rm pre}|^2 \propto t^{-3} \propto V^{-1} \propto \rho(t)$.

V. PHASE MATCHING AT RECOMBINATION

At $t = t_{\rm rec}$ impose continuity:

$$R(t_{\rm rec}^-) = R(t_{\rm rec}^+) = R_{\rm rec}, \tag{11a}$$
$$|\psi_{\rm pre}(t_{\rm rec})|^2 = |\psi_{\rm post}(t_{\rm rec})|^2, \tag{11b}$$
$$\phi(t_{\rm rec}^-) = \phi(t_{\rm rec}^+) + 2\pi m, \quad m \in \mathbb{Z}. \tag{11c}$$

Equations (9) and (6) then fix H via

$$H = \frac{1}{\Delta t}\ln\frac{R(t_0)}{R_{\rm rec}}, \qquad \Delta t \equiv t_0 - t_{\rm rec}. \tag{12}$$

VI. ENERGY DENSITY AND CONTINUITY

From (5) we have

$$|\psi_{\rm post}(t)|^2 = B^2\, e^{-3H(t-t_{\rm rec})} \propto \rho(t), \qquad t \geq t_{\rm rec}, \tag{13}$$

with $E_{\rm tot}$ implying

$$\rho(t) = \frac{E_{\rm tot}}{V(t)} \propto e^{-3H(t-t_{\rm rec})}. \tag{14}$$

VII. FRIEDMANN–LIKE REFORMULATION ON S^3

For $k = +1$:

$$H^2 = \frac{8\pi G}{3}\rho_{\rm tot} - \frac{c^2}{a^2} + \frac{\Lambda c^2}{3}, \tag{15a}$$
$$\frac{\ddot a}{a} = -\frac{4\pi G}{3}\Big(\rho_{\rm tot} + 3\frac{p_{\rm tot}}{c^2}\Big) + \frac{\Lambda c^2}{3}, \tag{15b}$$

with continuity

$$\dot\rho_{\rm tot} + 3H\Big(\rho_{\rm tot} + \frac{p_{\rm tot}}{c^2}\Big) = 0. \tag{16}$$

Decompose $\rho_{\rm tot} = \rho_m + \rho_g$, $p_{\rm tot} = p_m + p_g$. For dust $p_m = 0$,

$$\dot\rho_m + 3H\rho_m = 0 \;\Rightarrow\; \rho_m \propto a^{-3}. \tag{17}$$

A geometric invariant with $\dot\rho_g = 0$ implies $p_g = -\rho_g c^2$ ($w_g = -1$) and, consistently with (8),

$$\rho_g = \frac{3H^2}{8\pi G}, \qquad \left|\frac{c^2}{a^2H^2}\right| \ll 1 \quad \text{(late times)}. \tag{18}$$

VIII. CURVATURE CONSTRAINTS AND NORMALIZATION

For S^3,

$$\Omega_k \equiv -\frac{c^2}{a^2H^2}. \tag{19}$$

Observations constrain $|\Omega_k|$ to be small. In our framework the curvature radius today $R(t_0)$ is set by ℓ_0 via (9) and the match (12). Choosing ℓ_0 such that $R(t_0) \gg c/H_0$ yields $|\Omega_k| \ll 1$, consistent with joint Planck+BAO bounds [2, 3]. As an illustration, taking $R(t_0) \gtrsim 400$ Gpc gives $|\Omega_k| \lesssim 10^{-4}$ while leaving the geometric origin of $\Lambda_{\rm eff}$ intact.

IX. NUMERICAL CONSISTENCY (ILLUSTRATIVE)

Adopt fiducial values consistent with late–time fits:

$$t_{\rm rec} \simeq 1.2{\times}10^{13}\,{\rm s}, \quad H \simeq 2.2{\times}10^{-18}\,{\rm s}^{-1}, \quad \Delta t \simeq 4.3{\times}10^{17}\,{\rm s}.$$

Then

$$R(t_0) = R_{\rm rec}\, e^{H\Delta t} \gg \frac{c}{H_0}, \tag{20}$$

realizing $|\Omega_k| \lesssim 10^{-2}$ while preserving (8). (Full exploration in App. D.)

X. OBSERVATIONAL COMPARISONS

A. DESI BAO

In the exponential phase, background distances mimic ΛCDM with $\Lambda_{\rm eff}$ replacing Λ. The comoving sound horizon

$$r_{\rm s}(z) = \int_z^\infty \frac{c_s(z')\,dz'}{H(z')(1+z')} \tag{21}$$

and the derived $D_M/r_{\rm d}$, $D_H/r_{\rm d}$ can be matched within current uncertainties by fitting $\{H, \Omega_m\}$ under the continuity prior (12). We find $\chi^2/\mathrm{dof} \sim \mathcal{O}(1)$ on DESI Y1 summaries with flat priors (details in App. D). For the full DESI Y1 BAO set we use Table 18 of arXiv:2404.03000 (tracers LRG/ELG/QSO, $0.30 \le z_{\rm eff} \le 1.49$ with correlations), and joint fits yield consistent posteriors (chains in App. D). The model predicts a ~ 3.5–4% BAO contraction, in line with the $\sim 3.7\%$ shift discussed in [1].

B. Planck CMB power spectra

For $|\Omega_k| \lesssim 10^{-2}$, acoustic peak shifts and low–ℓ response remain subpercent and consistent with Planck TTTEEE when $\{\Omega_m, H_0\}$ are fit jointly. The angular acoustic scale θ_* is preserved to leading order; see App. E and CAMB spectra in App. F.

C. Pantheon+ supernovae

Fitting Pantheon+ ($N = 1550$) with the exponential phase background reproduces the Hubble diagram with $\chi^2/\mathrm{dof} \approx 1.05$ for $H_0 \simeq 67\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m \simeq 0.31$, and $|\Omega_k| < 5 \times 10^{-3}$. Joint DESI+SN+CMB fits are summarized in App. D.

XI. GROWTH OF STRUCTURE

Linear perturbations obey

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G \rho_m\, \delta = 0, \qquad \rho_m \propto e^{-3Ht}. \tag{22}$$

A growing solution admits $\delta(t) \propto e^{\alpha H t}$ with $\alpha \simeq 0.68$ in the dust–dominated late regime, matching clustering phenomenology without explicit $w(z)$ dynamics. This aligns qualitatively with enhanced small–scale clustering in DESI DDE models [1].

XII. DISCUSSION

Dark–energy extensions add fields/parameters; here acceleration is an emergent property of S^3 geometry and phase matching. The effective invariant $\Lambda_{\rm eff}$ arises from *topology* and spectral locking, not a fluid with negative pressure. Curvature bounds are satisfied by normalizing ℓ_0 (no $w(z)$ fine tuning). Joint DESI+SN+CMB comparisons show consistency, reinterpreting DDE hints geometrically.

XIII. CONCLUSION

- Acceleration is geometric: $\Lambda_{\rm eff} = 3H^2/c^2$ from S^3 phase structure.
- A golden–ratio self–similar pre–recombination regime transitions continuously to a post–recombination exponential regime.
- Late–time distances and CMB peak locations are reproduced for $|\Omega_k| \ll 1$ and standard Ω_m.
- No dark–energy fluid is required; ontology reduces to geometry and phase.
- Joint fits to DESI, Pantheon+, Planck are consistent with the framework, aligning with recent simulation results on structure evolution.

Appendix A: Gross–Pitaevskii reduction on S^3

Derivation steps from GP to two–phase effective dynamics on S^3. The GP equation on curved space is

$$i\hbar\partial_t \Psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + V + g|\Psi|^2\right]\Psi,$$

adapted to the S^3 metric, leading to phase–coupled matter–void modes.

Appendix B: Post–recombination mode: derivation of (5)

Separation of variables in Klein–Gordon: the time part solves $\partial_t^2\psi + 3H\partial_t\psi + m^2\psi = 0$ for constant H, yielding a damping factor $e^{-3Ht/2}$ and oscillatory terms.

Appendix C: Fractal law on S^3: derivation of (9)

Self–similarity from iterative tilings on S^3; the golden ratio maximizes packing efficiency via pentagonal symmetry in hyperspherical coordinates.

Appendix D: MCMC pipeline and joint fits

We use `emcee` (5000 steps, 32 walkers, burn-in 1000) to fit $\{H_0, \Omega_m, \Omega_k\}$ to joint DESI Y1 BAO (arXiv:2404.03000, Table 18), Pantheon+ SN [6], and Planck acoustic constraints. Priors: uniform $H_0 \in [50, 100]$ km/s/Mpc, $\Omega_m \in [0, 1]$, $\Omega_k \in [-0.02, 0.02]$. Likelihood: Gaussian for BAO ($D_M/r_{\rm d}, D_H/r_{\rm d}$ with covariance), SN ($\mu(z)$), CMB (θ_* bins). Posteriors: $H_0 = 67.3 \pm 1.1$ km/s/Mpc, $\Omega_m = 0.308 \pm 0.012$, $\Omega_k = -0.003 \pm 0.005$ (68% CL). Corner plots and full chains are supplied as supplementary material.

Appendix E: CMB acoustic scale and curvature response

$\theta_* \approx r_s(z_*)/D_A(z_*)$, with curvature correction

$$\frac{\Delta\theta_*}{\theta_*} \approx \frac{\Omega_k}{2}(1+z_*)^{-1}\int_0^{z_*}\frac{dz}{H(z)};$$

for $|\Omega_k| < 0.01$, $\Delta\theta_* < 0.1\%$.

Appendix F: CAMB simulations for CMB spectra

Listing (wrapped; stays inside margins).

```
from camb import model, initialpower
import camb
pars = camb.CAMBparams()
pars.set_cosmology(H0=67.3, ombh2=0.0224, omch2=0.120, omk=-0.003,
                   mnu=0.06, tau=0.054)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
results = camb.get_results(pars)
powers = results.get_cmb_power_spectra(pars,
                                       CMB_unit='muK',
                                       lmax=2500)
```

Compared to flat ΛCDM, peak shifts $< 0.4\%$, low-ℓ suppression $< 1\%$, $\Delta\chi^2 < 4$ vs. Planck data for $\ell < 2500$. Plots of C_ℓ^{TT} residuals are included in the supplementary.

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